

Independent supervision channel - Experimental proposal

Experimental architecture with physically separate communication path for the TAS Plf. NMR Supervision channel. State from 25 Aug 2026

The purpose is **only to evaluate technical feasibility** and provide input data for decision whether this architectural variant will be used in DIL-CU. Operational costs, design, cost of ownership is out of scope of this technical feasibility.

Motivation

Elimination of the split-brain hazard probability caused by double-fault of core routers. When core routers fails, it is not an immediate hazard. The dangerous part is during network re-integration when suddenly two masters might emerge. This requires special maintenance procedure in case of such double-fault. It cannot be fully ruled out that routers might spontaneously reboot due to a random fault. Furthermore, core routers are COTS devices with Basic Integrity, thus any safety-relevant assumption about their functionality and availability shall be taken with caution.

Hypothesis

Establishing physically separate (independent) communication path for the Supervision channel might improve the safety and prevent hazards caused by network re-integration and elimination of specific operational procedures or reducing the procedure in case of double-fault on core routers.

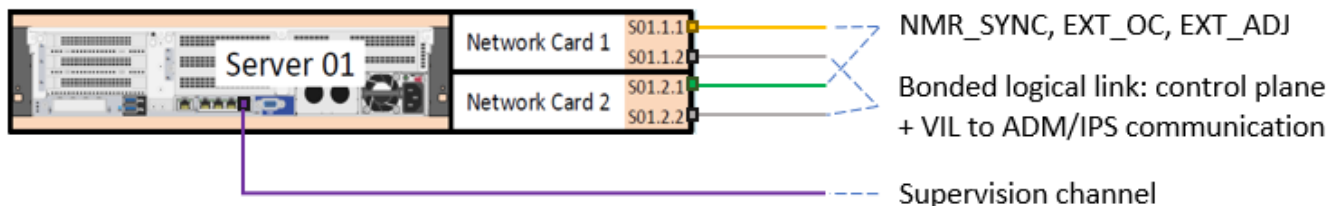
Proposal

Port allocation on server blades

Each server blade HP Proliant DL345 has 3 NICs:

- 2x NICs with 2x 10Gbps ports
- 1x NIC with 4x 1Gbps ports

Proposed physical port allocation:



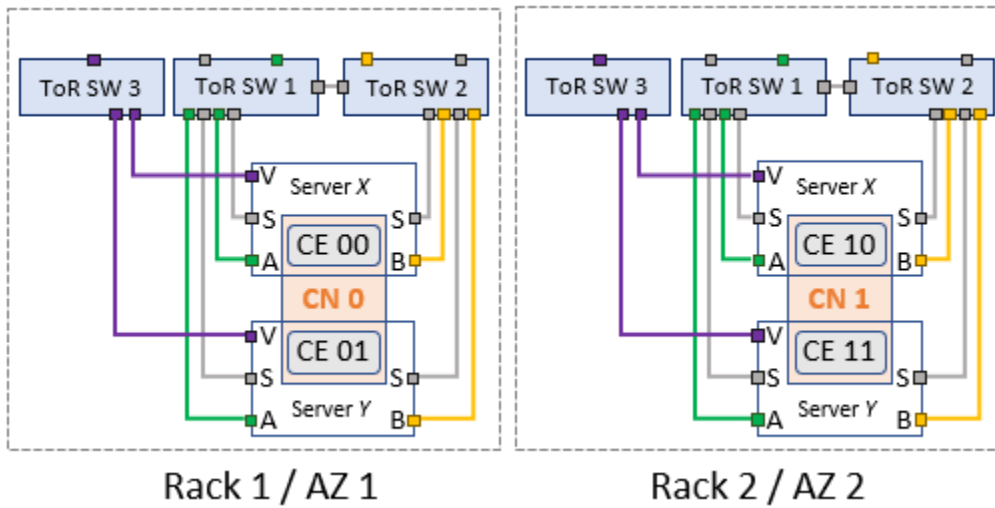
- 2x10Gbps cards:
 - 2 ports from each 10Gbps NIC (yellow and green) for independent physical links used for redundant networks where applications require two independent physical channels (NMR synchronization, communication with OCs, communication with adjacent systems etc.)
 - Remaining 2 NIC ports form so called **bonded logical link** (grey, S) – for control plane, and service (IPS) networks (VIL to ADM/IPS communication). The list of interfaces mapped to this network can be found in the [List of VIL interfaces and qualities](#)
- 4x1Gbps card:
 - 1 port exclusively only for TAS Plf Supervision channel (purple color)

Cabinet topology

Each cabinet/rack has 3 ToR switches:

- ToR SW 1 and ToR SW 2: For all interfaces from server blades except of the supervision channel interface
 - To keep physically independent communication path for TAS Plf NMR traffic ToR SW1 is exclusive only for the "sync channel A" (green color) and SW2 for "sync channel B" (yellow color)
 - SW1 and SW2 are connected via ICL (inter-chassis link). ToR SW3 is standalone, no ICL links
 - ServiceLink (gray color) which is for the control plane and all VIL to AMD/IPS traffic: forms a bonded link with both ToR SW1 and SW2.

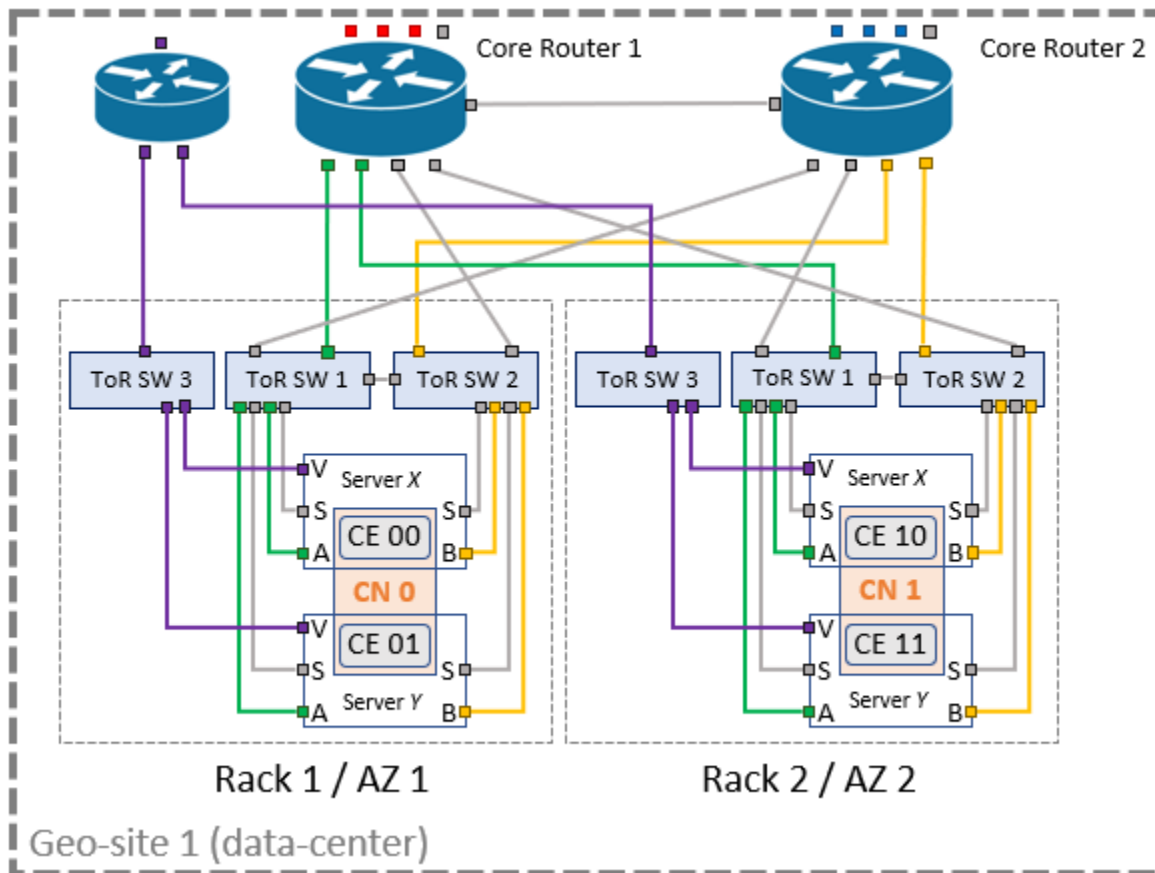
- ToR SW 3: Exclusively only for the supervision channel



Data-center topology

The proposal requires three routers:

- Core router 1 and 2 (NOKIA 7750 SR-2s):
 - To hold the independent communication paths requirement, NMR "sync channel A" from all racks/AZs goes to Core Router 1. NMR "sync channel B" is linked to Core Router 2.
 - ServiceLink goes to both core routers via own dedicated links from ToR switches and links are crossed: link from ToR SW 1 goes to CR 2, link from SW2 goes to CR1.
 - This cross link eliminates partitioned network in case of double fault with CR1 and ToR SW2 (or vice versa).
- "Mini" router (switch NOKIA 7250 IXR-e, aka "small" because it has MACSec):
 - Supervision channel (purple color) has own dedicated "mini" router"
 - Technically it is possible to avoid ToR SW 3 and connect Server blades directly to this mini router. It is up to the cost optimization.



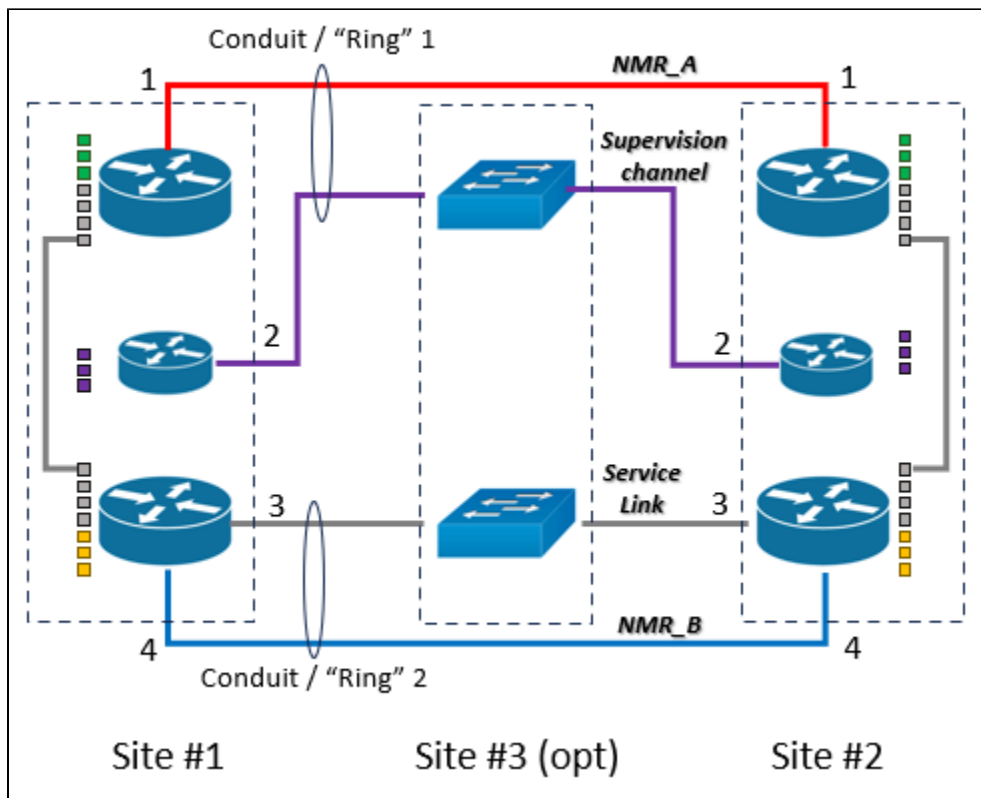
Interconnection of geo-sites

For geo-site interconnection are available four links where 2 links goes directly to the second data center and another pair goes via third location (Wels)

- Link 1 and 4: Proposed interconnection must adhere to the requirement of independent communication paths, thus 2 links out of 4 are exclusively dedicated for NRM sync channels A and B (red and blue color).
 - To minimize latency, links 1 and 4 go directly to another data-center and are not routed via third location
- Link 2: Dedicated link only for the supervision channel. This link is routed via Wels.
- Link 3: Geo-interconnection for the ServiceLink, i.e. control plane and VIL to ADM/IPS communication. This link is routed via Wels.

In the third location (Wels), routers do not have any ICL link and do not "mix" traffic between ServiceLink and Supervision channel. If one link or device fails, there is no dynamic re-routing via alternative path. In this configuration we need all 4 links to break to create a safety hazard.

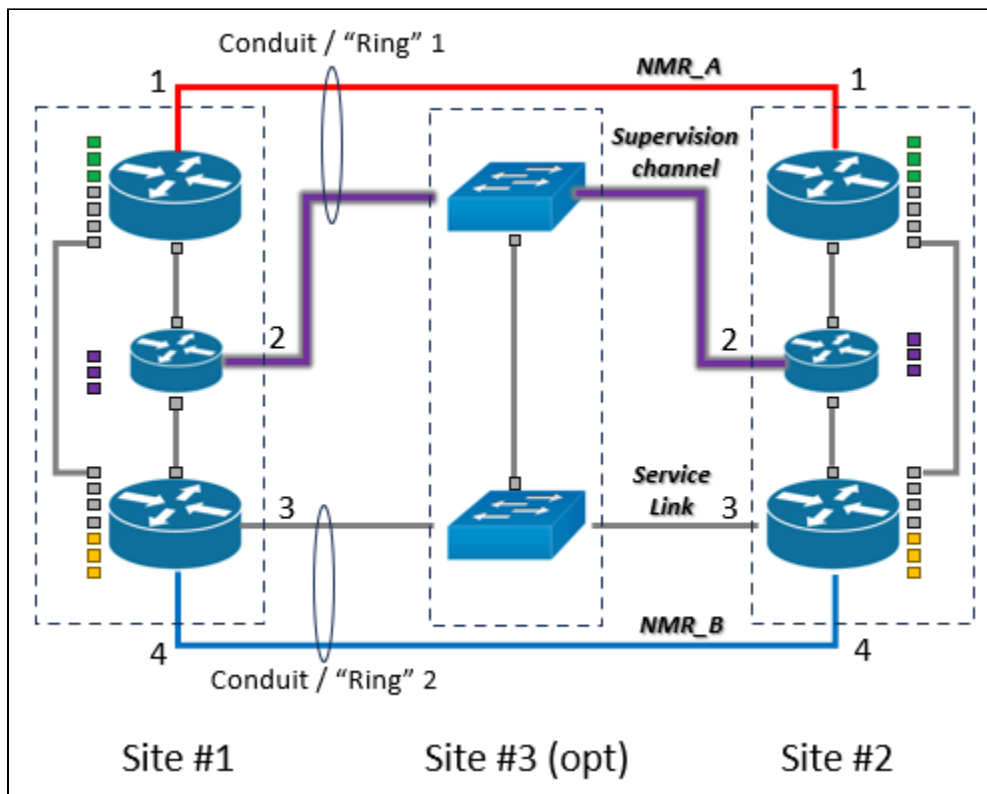
Diagram of the proposal:



The disadvantage of the above proposal is lower availability of ServiceLink because it fully relies on one link instead of 2 (or more). This can be improved with the following "optimization":

- Link for the Supervision channel can be used as a backup route in case of the ServiceLink failure.
- QoS for Supervision channel must be higher than the backup route
- Traffic from the Supervision channel shall not be routed via alternative path in case of failure - no switch over.
- As soon as the link restored, ServiceLink is re-routed back to the primary path

The improved proposal:



Analysis (safety and availability)

Comparison with the initial proposal without 3rd link (reference proposal):

- Inside the data center:
 - The third channel **does not improve safety inside the data center** wrt the reference proposal. The reference proposal already eliminates hazards caused by double faults. Furthermore, double fault of ToR SW1 and SW2 causes disintegration of CN in the affected rack.
 - The third channel **does improve EL-CORE availability** in case of basic-redundancy. If both core routers are down, then NMR and ServiceLink networks are partitioned. Slave CN will try to become a new master but due to a Supervision channel it will be prevented to do so. If there is no independent supervision channel, the slave would become master - there would be 2 masters but fully isolated also from the field - thus, no safety risk. However, upon re-integration very likely whole CG would be rebooted and the availability is impacted. With the proposed independent supervision channel this would not happen: Master will remain in its role, Slave will end in cyclic reboot. During the integration the master will continue operation and slave will re-integrate.
 - Impact on improvement of availability of ECMUX is to be further investigated.
- There is a hazard (partitioned network) if failure of both core routers and also the "mini" router happens altogether. This is extremely unlikely, below SIL-4 hazard rate. Exposure time in case of both core routers fail: to be calculated.
 - Third independent channel **does improve safety in case of double core routers failure** - prevents electing a new master:
 - In case both core-routers CR1/CR2 are permanently down (common cause or independent faults)
 - Spontaneous reboot of core routers CR1 and CR2 due to various internal faults
 - One is in down due to maintenance and the second goes to reboot or become temporarily (hundreds of milliseconds) unavailable

Channel monitoring

It is critical to monitor whether the channel is operational. TAS Plf offers the following mechanism:

On the receiving side there is a NOTICE level log saying "adsv connection up(down)" when the additional supervision channel transitions from not working to working and vice versa. To determine whether it is working it compares the time of the last tick and the time of the last advs message, so if this difference is bigger(equal) than the master switch timeout then the advs channel is not working. We should get a "adsv connection up" once everything is started, and a "adsv connection down" if the channel stops receiving messages, or messages are outdated.

MoM with Robert Holzkecht on 24.08.2026

Summary:

Proposal is technically feasible and eliminates hazard in case of the double failure of core routers.

Nevertheless it costs more to build and maintain (see below).

It has to be decided by management whether additional costs related to this HW solution is acceptable, or if another solution (e.g. SW based - if possible) is cheaper.

Pros:

- Eliminates the split-brain hazard in case of double core routers failure
- Prevents split-brain in transient situations such as:
 - Automatic restart of core routers
 - Outage of a core router or ToR + flipping Ethernet port on the other network redundancy in case of EC-MUX
 - One is in down due to maintenance and the second goes to reboot or become temporarily (hundreds of milliseconds) unavailable
- No need for a change in SW
- Solution at the infrastructure level

Cons:

- Higher maintenance costs
 - Additional cost on Hitachi to maintain different IPS topologies for different product
 - introduction of an additional MACsec capable router in the network rack, leading to:
 - since the 7250 IXR-e small is not sufficient (according to our concept we need 50 Gbit/s), a new device has to be evaluated
 - Has to be evaluated together with Nokia (coordinated together with Lukas Dietrich); an option, if supporting MACsec, might be the 7250 IXR-e2
 - this leads to redesign and requalification of the rack and hardware combination
 - lowering the overall availability of this network path (minimal but still present - one device more which could break down)
 - additional costs, raising the customers LCC (not agreed and covered by the tender)
-